Pac-Man Classic Game (Java Implementation)

Project Name: **Pac-Man Classic**

Project Objectives:

The main objectives of this project are:

- **To apply Object-Oriented Programming (OOP) concepts** such as Inheritance, Encapsulation, Polymorphism, and Abstraction in a practical game development project.
- **To develop a graphical game interface** using Java AWT/Swing and implement essential game mechanics such as player movement, collision detection, scoring, and level progression.
- **To implement real-time event handling and game logic** using keyboard inputs, a structured game loop, and basic Artificial Intelligence (AI) for ghost movement.

Project Outcome:

The project will result in a functional **Pac-Man-style arcade game developed using Java**. Players will be able to control Pac-Man, collect food pellets, avoid ghosts, earn scores, and progress through different levels.

From a societal and educational perspective, this project will help students understand how theoretical concepts of **Object-Oriented Programming, Data Structures, Algorithms, and Event Handling** can be applied to develop a practical software application. It will also provide fundamental knowledge of game development, logical problem-solving, and software design, which can be useful for future programming and software engineering projects.

Project Features:

The following features will be implemented in the project:

1. **Tile-Based Map Loading**
   A 2D array will be used to design and manage the game grid, including walls, paths, and food pellets.
2. **Player Movement**
   Pac-Man will be controlled in real time using keyboard arrow keys.

3. **Collision Detection**

   The game will detect collisions between Pac-Man, walls, ghosts, and other game objects.

4. **Ghost AI**

   Ghosts will use a basic random movement algorithm to create challenges for the player.

5. **Scoring System**

   Players will earn points by consuming food pellets throughout the game.

6. **Life System**

   Pac-Man will have 3 lives, and the game will track the player's remaining lives.

7. **Game Loop**

   A structured game loop will continuously update the game state and render the game for smooth gameplay.

8. **Game Over Screen**

   The system will display appropriate win or loss messages when the game ends.

9. **Level Progression**

   After completing a level, the game map will be reloaded or updated to allow the player to continue.

10. **Restart Feature**

    Players will be able to restart the game and reset the game state after losing or completing a game.

Project Timeline:

Timeline	Tasks
Week 1	Project planning, requirement analysis, Java environment setup, and initial system design
Week 2	Design and implementation of the tile-based map using 2D arrays
Week 3	Develop the core graphical interface using JFrame, JPanel, and Java AWT/Swing
Week 4	Implement Pac-Man character movement and keyboard event handling
Week 5	Develop and test collision detection between Pac-Man, walls, and game objects
Week 6	Implement ghost movement and basic Ghost AI
Week 7	Implement food consumption, scoring system, and score display
Week 8	Implement life mechanics and optimize the main game loop
Week 9	Implement restart functionality, game-over screen, winning conditions, and level progression
Week 10	Complete testing, debugging, performance improvements, documentation, and final project preparation